

GaussCluster: A 3D Gaussian Mixture Model for Open Cluster Membership

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Abstract

Context. Isolating true dynamical members from the galactic background is essential for characterizing open clusters. In highly obscured environments like the Cygnus X region, spatial density algorithms struggle, often sweeping up heavy background contamination.

Aims. We develop a two-stage methodology which couples blind spatial-density searches with a probabilistic astrometric extraction, enabling the recovery of a cluster’s core even under extreme reddening conditions.

Methods. We present **GaussCluster**, a 3D Gaussian Mixture Model pipeline utilizing high-precision Gaia DR3 proper motions and parallaxes. To evaluate the pipeline’s efficiency, we applied it to the heavily obscured young cluster Berkeley 87 and a less complex control target, NGC 957. The derived astrometric populations were then cross-matched with theoretical PARSEC isochrones to establish their fundamental parameters.

Results. We isolated 70 highly probable ($P \geq 0.9$) astrometric members in Berkeley 87 and 180 in NGC 957, by anchoring our model on the seed cores identified by the spatial algorithm, STORM, within a 15-arcminute boundary. We derived a distance of $\sim 1.8 \pm 0.13$ kpc for Berkeley 87, and physical parameters for NGC 957 (10 Myr, 2.2 kpc), consistent with the literature. The 3D filtering revealed a photometric sequence yielding an isochrone age of ~ 4.5 Myr for Berkeley 87, aligning with the ≤ 5 Myr physical limits dictated by the presence of post-main sequence stars.

Conclusions. The purely astrometric, probabilistic workflow of **GaussCluster** provides an alternative approach to overcoming the observational biases caused by clumpy interstellar extinction and spatial overlapping. This methodology provides a foundation for building accurate catalogues of young stellar populations in complex environments.

1. Introduction

1.1. Overview

The accurate determination of open cluster parameters such as age, distance, and initial mass function is central to studies of galactic structure and stellar evolution [1]. However, extracting a clean catalogue of physically bound cluster members from the dense galactic background is a complex challenge, particularly in obscured regions of the Milky Way [2, 3]. This paper introduces **GaussCluster**, a 3D astrometric modelling pipeline designed to address this membership problem using high-precision data from the Gaia mission [4].

1.2. Open Clusters as Astrophysical Laboratories

Open clusters are widely recognised as essential tools in observational astronomy [2]. As gravitationally bound groups of stars born from the same giant molecular cloud at approximately the same epoch and distance, they provide astronomers with a natural laboratory for studying stellar evolution [1], particularly in their early embedded phases [5]. While theoretical isochrone fits to a cluster’s Colour-Magnitude Diagram (CMD) are traditionally used to constrain age, distance, and reddening [6], the advent of Gaia allows distance to be constrained astrometrically,

breaking photometric degeneracy. This makes open clusters valuable calibrators of stellar evolution timelines [7] and theoretical models [6], particularly those involving rotational mixing [8]. Comparing cluster observations to theoretical tracks like the PARSEC models provides a direct test of stellar interior physics, like convective overshooting, rotation, and mass loss [6].

Beyond the study of individual stars, the macroscopic spatial distribution and ages of open clusters across the galactic disc are useful in tracing spiral arm structures and understanding the history of star formation within the Milky Way [9, 10]. Cluster classification has helped map the Galaxy [11, 12], while modern surveys continue to trace spiral arm structures with extreme precision [13].

Young open clusters, with ages on the order of tens of millions of years, are important as they have not had sufficient time to dissolve into the galactic field [5]. Because they have not yet been dissolved by two-body relaxation and tidal stripping, they retain a population of massive, early-type (O and B) stars [1]. These stars evolve rapidly and inject substantial amounts of mechanical and radiative energy into the ISM via stellar winds and subsequent core-collapse supernovae, making them sensitive tracers of recent star formation activity [14]. The initial mass function (IMF) of these young clusters also provides valuable insight into the star formation process itself, helping

answer questions about whether stellar mass distributions are universal across different galactic environments [15].

1.3. The Impact of High-Precision Astrometry

Deriving accurate cluster parameters depends on the observer's ability to correctly separate physically bound cluster members from foreground and background field stars along the line of sight [16, 17]. Before the onset of high precision astrometric surveys, membership was assigned photometrically [18, 19]. This involved plotting all stars in a given field on a CMD and statistically subtracting field stars based on a theoretical isochrone or empirical main sequence [20]. Such methods have been standard practice for decades [19], although they are susceptible to foreground contamination [21, 22].

In crowded or reddened regions along the galactic plane, reddened foreground stars or luminous background objects can easily appear as obscured cluster members [22, 2]. The accurate proper motions and parallaxes provided by Gaia Data Release 2 (DR2) [23, 24] and Data Release 3 (DR3) [25] allow cluster membership to be assigned astrometrically, largely independent of a cluster's photometric properties [4].

This has started a revolution in cluster science, as astronomers have used Gaia astrometrics to revise cluster parameters, discover entirely new clusters hidden in the galactic plane, and discard objects obtained from earlier photometric-only studies [26]. Recent works including density-based spatial clustering algorithms like HDBSCAN [27], machine learning models such as Random Forests [28], and probabilistic clustering models like UPMASK [29] which was adapted for Gaia, all use this data. Together, these projects have refined parameters for specific clusters [30, 31], resolved complex star-forming regions [3], and generated large scale, all-sky cluster catalogues [13]. However, massive homogeneous catalogues rely on global algorithmic settings. While these global settings excel at discovering average clusters across the whole sky, they struggle or fail entirely when confronted with peculiar, extreme differential reddening environments like the Cygnus X region [27, 3].

1.4. The Cygnus X Region and the ON2 Complex

To fully understand the observational challenges associated with Berkeley 87 (hereafter: Be87), it is important to contextualise its location within the Milky Way. The cluster resides deep within the Cygnus X region, one of the most active star-forming regions in the local galactic neighbourhood [32]. Spanning several degrees across the sky, Cygnus X contains millions of solar masses of molecular gas and dust, driving star formation that powers numerous HII regions and OB stars [3]. Within this area resides the ON2 star-forming region, characterised by variable interstellar extinction [33]. The molecular clouds in this specific region absorb optical light, making observations at shorter optical wavelengths challenging [34].

Because the cluster is still physically present within, or lying immediately behind its natal molecular cloud, the dust distribution is clumpy [5, 3]. This clumpy structure is what generates the severe differential reddening that affects all optical photometric studies of the cluster, as photons from adjacent stars must travel different column densities of interstellar material before reaching Earth [3].

1.5. Be87: A Complex and Obscured Target



Figure 1: H α image of Be87 using the LCO 0.4m telescope, with clear dust obscuration visible

The first complete observational characterisation of Be87 (Figure 1) was carried out by Turner & Forbes (1982) [35], who identified it as host to several highly evolved stars, which includes the emission-line variable V439 Cyg and the oxygen-rich Wolf-Rayet star WR 142. The presence of WR 142 is diagnostic as it indicates that Be87 must have formed massive progenitor stars ($> 40M_{\odot}$) [36], placing constraints on the cluster's upper age limit, as such stars typically undergo core-collapse supernovae within ~ 5 million years [37].

High-energy X-ray observations have also provided a characterisation of this stellar population. Observations by the Chandra X-ray Observatory confirmed the membership of WR 142 and several OB-type stars while revealing hot thermal plasma emission generated by colliding stellar winds, typical of dense, young clusters [36], providing further evidence of the cluster's extreme environment [38].

More recently, a study by de la Fuente et al. (2021) [3] compiled a multi wavelength survey of the Be87/ON2 field, providing a Gaia EDR3-based distance determination alongside a catalogue of confirmed cluster members with known spectral types. This work addressed the difficulty of handling overlapping young stellar populations along this crowded line of sight [3], a task which previous spatial

algorithms had struggled to map accurately due to the clumpy dust distribution [27].

This project aims to directly address this spatial-overlapping limitation, by establishing fundamental parameters through Gaussian astrometric membership selection. Evaluating highly obscured clusters requires confidence in the methodology, so this study incorporates an independent check on a separate open cluster, NGC 957. By comparing this purely astrometric model against a spatial density-based algorithm on the control target (NGC 957), this study aims to demonstrate how a Gaussian mixture model can filter out spatial contamination before deploying the pipeline to resolve the highly obscured Be87.

The structure of this paper is as follows: Section 2 outlines the foundational methodology and astrometric data acquisition. Section 3 details the architecture of the **GaussCluster** pipeline, Section 4 presents the astrometric recovery of Be87 and NGC 957, and details the derivation of parameters via isochrone fitting. Section 5 provides a summary of the findings, and Section 6 concludes with a brief discussion of future works.

2. Methodology

The methodology implemented in this study is centred on the utilization of precision astrometry from the Gaia mission to isolate true cluster members from the galactic background.

2.1. Astrometric Data Acquisition and Filtering

To determine the structure of Be87, astrometric data was retrieved from the third data release of the Gaia mission (Gaia DR3) [4] via the **astroquery** Python package [39]. A circular cone search was performed centred on the accepted cluster coordinates ($\alpha = 20^{\text{h}}21^{\text{m}}35.0^{\text{s}}$, $\delta = +37^{\circ}25'59''$, J2000)[40], out to a radius of 0.5 degrees (30 arcminutes), to ensure a sufficient sample of background stars for the subsequent field density models. However, the final spatial boundary of the extracted cluster was restricted to 15 arcminutes to capture the inner core and primary halo while mitigating extreme field contamination. This spatial restriction is consistent with the typical structural extents and tidal radii of open clusters residing within the galactic disc [41]. While stars in the extended outer halo may have been ejected from the core via dynamical interactions [42, 43, 44], limiting the radius to 15 arcminutes ensures the subsequent models are not overwhelmed by the non-uniform background field.

To ensure high astrometric accuracy before applying any statistical models, the initial dataset was filtered. The Gaia catalogue contains ~ 27000 sources in this region, but many suffer from poor astrometric solutions [4, 25]. Only targets where the measurement errors in proper motion ($\epsilon_{\mu_{\alpha*}}$, $\epsilon_{\mu_{\delta}}$) were small (≤ 0.5 mas/yr) were retained. The catalogue is then filtered based on the Renormalised Unit Weight Error (RUWE). A high RUWE typically indicates a poor fit to the single-star astrometric model, often

caused by unresolved binary companions perturbing the photocenter [45]. A $\text{RUWE} < 1.4$ cutoff was applied to ensure the astrometric solutions were consistent [23]. This pre-processing step removed inaccurate measurements, unresolved close binaries, and faint background stars with uncertain astrometrics that could artificially inflate the calculated cluster velocity dispersions [46].

2.2. Astrometric Membership Determination

A limitation of traditional spatial and photometric studies of Be87 is the assumption that stars clumped together visually in the sky belong to the cluster [3]. To isolate dynamical members from the dense Cygnus X foreground and background, a probabilistic approach was adopted, expanding upon the bivariate mixture model framework initially formulated by Balaguer-Núñez et al. (1998) [16]. While the traditional model relies purely on 2D proper motions, this approach is susceptible to line-of-sight contamination. Therefore, the model was adapted into a 3D Gaussian Mixture Model (3GMM) by incorporating Gaia parallax (ϖ) as a third astrometric dimension, extending the model into a fully three-dimensional spatial-astrometric framework.

In this expanded framework, the distribution of all stars is modelled as a multivariate normal distribution defined by a data vector $\mathbf{x} = (\mu_{\alpha*}, \mu_{\delta}, \varpi)^T$. For a given star with corresponding observational covariance Σ_{ϵ} , the probability density function for the cluster, Φ_c , is generalized as [47]:

$$\Phi_c = \frac{1}{(2\pi)^{3/2}|\Sigma_c + \Sigma_{\epsilon}|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T(\Sigma_c + \Sigma_{\epsilon})^{-1}(\mathbf{x} - \boldsymbol{\mu}_c) \right\} \quad (1)$$

where $\boldsymbol{\mu}_c$ represents the systemic 3D centroid of the cluster (proper motions and parallax) and Σ_c is its intrinsic dispersion matrix.

However, because the background field stars in the Cygnus plane are highly non-uniform and cannot be described by a single broad Gaussian, the field distribution (Φ_f) was modelled using a multi-dimensional Kernel Density Estimate (KDE) [48, 29]. By treating the spatial-astrometric field data as a sum of local Gaussian kernels, the KDE models the non-Gaussian structure of the field. For a given data point $\mathbf{x}$, the field probability density is evaluated as:

$$\Phi_f(\mathbf{x}) = \frac{1}{N_f} \sum_{i=1}^{N_f} K_{\mathbf{H}}(\mathbf{x} - \mathbf{x}_i) \quad (2)$$

where N_f is the number of field stars used to train the KDE, $K_{\mathbf{H}}$ is the multivariate Gaussian kernel, and $\mathbf{H}$ is the bandwidth matrix determining the smoothing scale (calculated via Scott's rule [49]). The final membership probability P for each star is then defined as the ratio of the cluster density to the total combined density (cluster plus field)[50, 17]:

$$P = \frac{n_c \cdot \Phi_c}{n_c \cdot \Phi_c + n_f \cdot \Phi_f} \quad (3)$$

where n_c and n_f represent the normalized fractions of cluster and field stars in the region, respectively, and $n_c + n_f = 1$. These parameters were solved using the empirical Gaia data, giving a converged catalogue of 3D astrometric members. A probability threshold of $P \geq 0.9$ was adopted for the derivation of parameters.

2.3. Photometric Validation via Gaia DR3

To validate the astrometric membership model, a CMD was constructed, using the native Gaia DR3 passbands (G , G_{BP} , G_{RP}) for all stars identified as highly probable members ($P \geq 0.9$). Rather than relying on external ground-based photometric catalogues, the use of Gaia photometry ensures homogeneity [51], allowing a direct visual confirmation that the astrometrically selected stars form a distinct, physical main sequence.

Theoretical stellar evolutionary tracks were generated from the PARSEC database [6] assuming solar metallicity ($Z = 0.0152$), which is appropriate for young galactic disc objects [52]. Rather than simultaneously attempting to fit age, distance, and reddening, an approach which frequently leads to degenerate parameter solutions [53], the distance modulus was anchored using the median zero-point-corrected Gaia DR3 parallax of the high-probability members [25], following the statistical parallax methodologies outlined by [24]. This calculation allowed us to treat distance as a fixed prior. By locking the vertical position of the isochrone, the remaining properties (age, reddening) could then be evaluated against the astrometrically selected population, breaking the observational degeneracy [8].

3. The GaussCluster Pipeline

Traditional open cluster studies often rely on manual, tailored code that is calibrated to a single target [2, 27]. Rather than writing target-specific analytical scripts, the 3D astrometric framework was engineered into a reusable Python Command Line Interface (CLI) tool, **GaussCluster**¹.

3.1. Pipeline Architecture

GaussCluster was designed to automate the process of separating intrinsic cluster astrometrics from background field contamination. The architecture of the software is built around a sequential pipeline:

1. **Data Acquisition & Filtering:** In this integrated pipeline, the spatial output generated by STORM [54] is ingested directly into **GaussCluster** [55] to act as the initial 'seed' members for the target cluster. The software interfaces directly with the Gaia DR3 archive via the **astroquery** API [39] to download the complete stellar field out to a customizable search radius, which the user defines based on the

expected tidal extent of their specific target. For the case of this paper, the radius is set to 0.5 degrees. To prevent astrometric noise or unresolved binary systems from artificially inflating the velocity dispersion, the software automatically purges any source with a $\text{RUWE} \geq 1.4$ [23] or proper motion errors exceeding 0.5 mas/yr.

2. **3D Astrometric Core Anchoring:** Unlike blind algorithms [54], **GaussCluster** anchors its first step by using a 3D sigma-clipping algorithm on the proper motions ($\mu_{\alpha*}, \mu_{\delta}$) and parallax (ϖ) of the ingested core members to calculate an initial estimate for the cluster's centroid. The cluster distribution is then modelled as a trivariate Gaussian.
3. **Non-Parametric Field KDE:** The software employs a Non-Parametric Kernel Density Estimator (KDE) to map the asymmetric astrometric density of the surrounding galactic field. This non-parametric approach to field modelling has become standard practice in probabilistic membership selection [29].
4. **Dynamic Bayesian Prior:** The algorithm evaluates the astrometric density of stars in the core and subtracts the expected KDE field density, dynamically calculating the local cluster density to establish a Bayesian prior (n_c). This helps reduce manual steering and allows the software to automatically discover the full physical scale of the cluster.
5. **Diagnostic Export:** Finally, the software extracts all stars above a threshold ($P \geq 0.9$) and generates a diagnostic dashboard, based strictly on the final 3GMM astrometric probabilities. This can be seen in Figure 2 which shows the output of **GaussCluster** when run on Be87.

3.2. Methodological Choices and Limitations

While the 3D astrometric Gaussian Mixture Model isolates cluster members in heavily obscured regions, the implementation of the algorithm required multiple methodological decisions to ensure stability and computational efficiency.

3.2.1. Decoupled Field Contamination Modeling

The field-star mixing fraction, n_f , was estimated independently of the KDE field density model. A geometric argument based on the excess source density beyond the core exclusion radius was used, assuming approximate local uniformity of field sources in normalised astrometric space [26, 56]. This separates the total number of expected background stars from the actual shape of their astrometric distribution, meaning the two are not required to be consistent. This two-stage decoupling approach is often utilised in astrometric cluster models to prevent local anomalies from artificially inflating the expected global field density [57].

¹The source code for the **GaussCluster** pipeline is publicly available at <https://github.com/raffey05-sudo/GaussCluster>

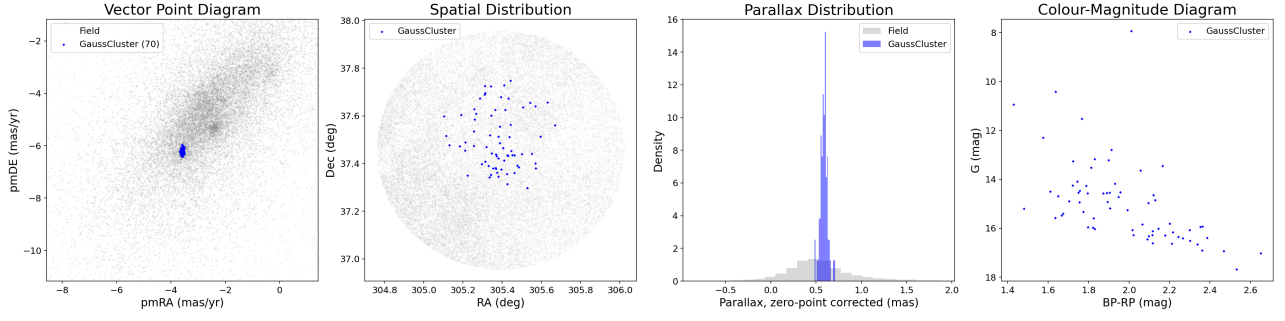


Figure 2: GaussCluster 4-panel diagnostic dashboard for Be87. The algorithm recovered 70 members (≥ 0.9), forming a tight Main Sequence.

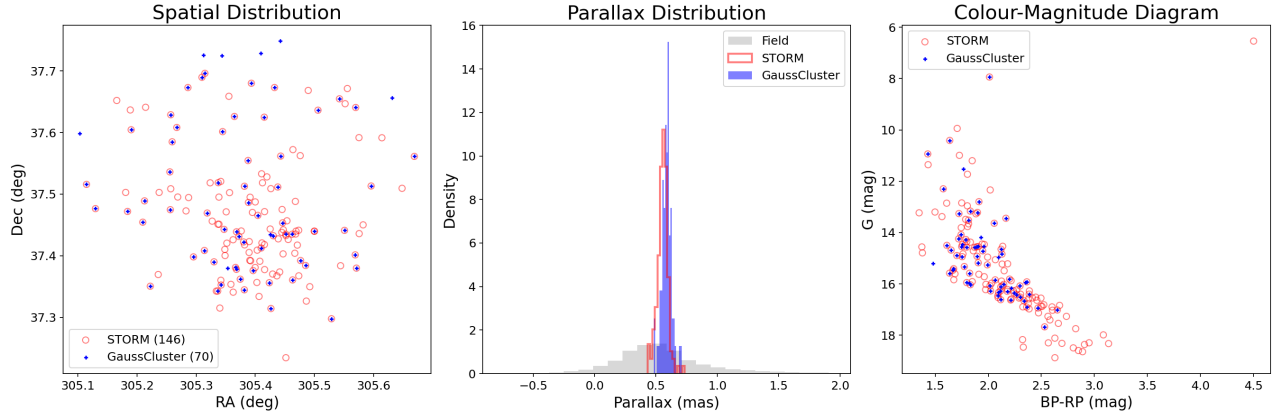


Figure 3: Spatial and parallax distribution comparison, along with a Colour-Magnitude Diagram for Be87. GaussCluster model (blue, 70 members) versus STORM (red, 146 members within 15 arcminutes).

3.2.2. Mahalanobis Distance Truncation

Standard E-M algorithms are sensitive to outlier contamination, a known limitation of Finite Mixture Models [58]. To ensure that these outliers do not artificially widen the cluster model during the iterative loop, the membership probability for any source lying beyond a Mahalanobis distance of 3σ was forced to zero [59, 29]. This is equivalent to fitting a truncated, rather than a full, trivariate Gaussian to the astrometric distribution at each step. While this introduces a downward bias in the recovered velocity and parallax dispersion by excluding high-dispersion tail members [60, 61], it guarantees algorithmic stability.

3.2.3. Static KDE Background Estimation

To prevent iterative instability during maximum likelihood estimation [47], the background field KDE was fit only once using the field sample defined by the initial core-based cluster estimate, rather than being continuously refit at each E-M iteration [48, 57]. This static implementation assumes that the core-field boundary does not shift enough during refinement to alter the field sample's composition. Given the stable convergence observed, this assumption is well supported as the astrometrically distinct nature of the initial STORM seed cores ensures that the field

selection remains representative throughout the model's execution.

4. Results and Discussion

The standard output generated by the GaussCluster pipeline is a 4-panel diagnostic dashboard, visualized in Figure 2 for Be87. Reading from left to right, the dashboard displays the isolated cluster's proper motion Vector Point Diagram (VPD), its spatial footprint (RA/Dec), its parallax distribution, and the resulting Colour-Magnitude Diagram (CMD). To evaluate the two-stage pipeline, this expanded GaussCluster output was analyzed in the context of the initial STORM core input and the recent catalogue published by Hunt & Reffert [27].

4.1. Astrometric Structure of Berkeley 87

Using the STORM extraction as the initial seed core, the application of the 3GMM yielded 70 highly probable members for Be87 at the $P \geq 0.9$ confidence threshold. The complete catalogue of these 70 core members, including their astrometric parameters and final membership probabilities, is provided in Appendix E. The converged 3GMM returned a systemic cluster centroid with intrinsic dispersions of $\mu_\alpha = -3.584 \pm 0.045$ mas/yr, $\mu_\delta = -6.237 \pm 0.101$

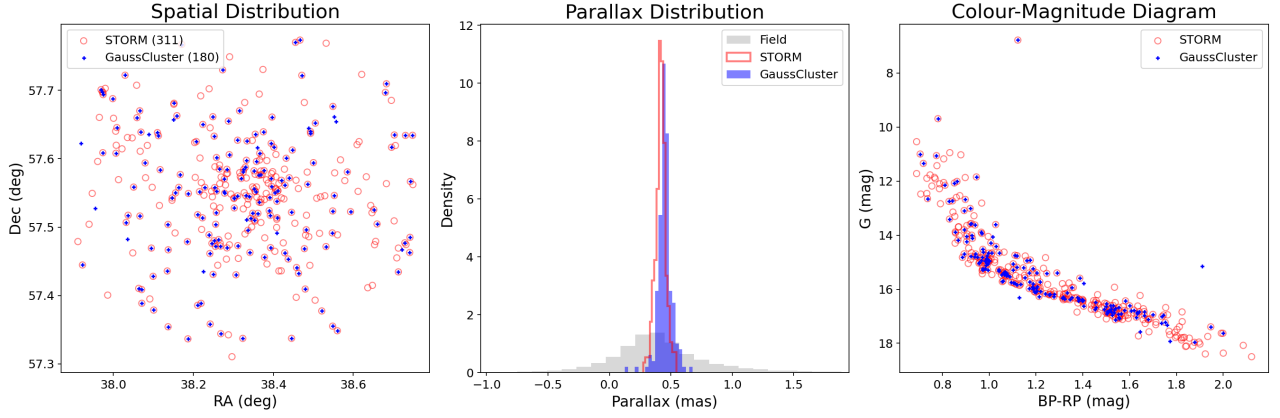


Figure 4: Spatial and parallax distribution comparison, along with a Colour-Magnitude Diagram for NGC 957. **GaussCluster** model (blue, 180 members) versus STORM (red, 311 members within 15 arcminutes).

mas/yr, and a median parallax of $\varpi = 0.555 \pm 0.040$ mas.

The difference in total membership size from STORM (146 members within 15 arcminutes) to GaussCluster (70 members) highlights the divergence between spatial-density and astrometric approaches. Density-based algorithms like HDBSCAN (the clustering engine behind STORM and Hunt 2023) rely on identifying continuous regions of high point density. In crowded galactic plane fields like Cygnus X, the dense background can bridge with the cluster, causing density algorithms to include larger numbers of field stars. For comparison, cross-matching our 70 astrometric members with the independent [27] catalogue yielded an intersection of 34 common stars (a 48.6% overlap with our core), further highlighting the divergence between probabilistic and spatial-density approaches in highly obscured fields.

By operating in three-dimensional astrometric space $(\mu_{\alpha*}, \mu_{\delta}, \varpi)$ and utilizing the full observational covariance matrix, **GaussCluster** applies a stricter astrometric constraint. It rejects spatial interlopers that do not share the cluster’s precise systemic motion. The inclusion of parallax as a third dimension ensures that even if a star resides within the spatial core, it is discarded if it does not share the same distance modulus. This demonstrates that while density models encompass a wider spatial footprint, probabilistic 3D models return a more conservative, astrometrically restricted physical population. To verify that this core reflects a bound structure, empirical King Profile models [62] were fitted to the spatial distributions of both clusters (see Appendix D).

Note that when comparing the resultant parallax distributions, as seen in both Figure 3 and Figure 4, a slight systematic offset is visible between **GaussCluster** and the literature catalogues. This occurs because the **GaussCluster** pipeline explicitly applies the Lindegren et al. (2021) [25] zero-point corrections to the native Gaia DR3 parallaxes, whereas the comparison catalogues rely on uncorrected values.

4.2. Methodology Validation: NGC 957

To ensure the pipeline was not biased by the spatial density and structural complexity of the Cygnus X field, a separate validation of **GaussCluster** was performed on the open cluster NGC 957. Ingesting the STORM core (311 sources) as the initial seed, **GaussCluster** extracted a final 3D population of 180 members, where 165 of the 180 (91.7%) are shared with the STORM sample. Cross-matching this output with the independent [27] catalogue gave an intersection of 134 common members.

Both the STORM and Hunt methodologies extracted significantly more total members for NGC 957 (311 and 252 respectively) than **GaussCluster**. Consistent with the results from Be87, these variations are attributable to clustering mechanics. Because both STORM and Hunt rely on spatial-density algorithms, they are prone to absorbing faint background sources that spatially overlap with the cluster. The 3GMM mathematically filters out these spatial interlopers, demonstrating that the pipeline consistently recovers a more astrometrically restricted physical sequence, regardless of the cluster’s environment.

4.3. Photometric Validation and Fundamental Parameters

Table 1: Fundamental parameters derived for NGC 957 and Be87 compared against literature values.

Parameter	NGC 957	Berkeley 87
<i>This Work</i>		
Age (Myr)	10.0	4.47
Distance (kpc)	2.2	1.80
<i>Literature Mean</i>		
Age (Myr)	10.0 ^a	2.7 ^b
Distance (kpc)	2.0 ^a	1.45 ^b

^aMean of [20] (10 Myr, 2.2 kpc) and [13] (10 Myr, 1.8 kpc).

^bMean of [35] (1-2 Myr, 1.2 kpc) and [13] (4 Myr, 1.7 kpc).

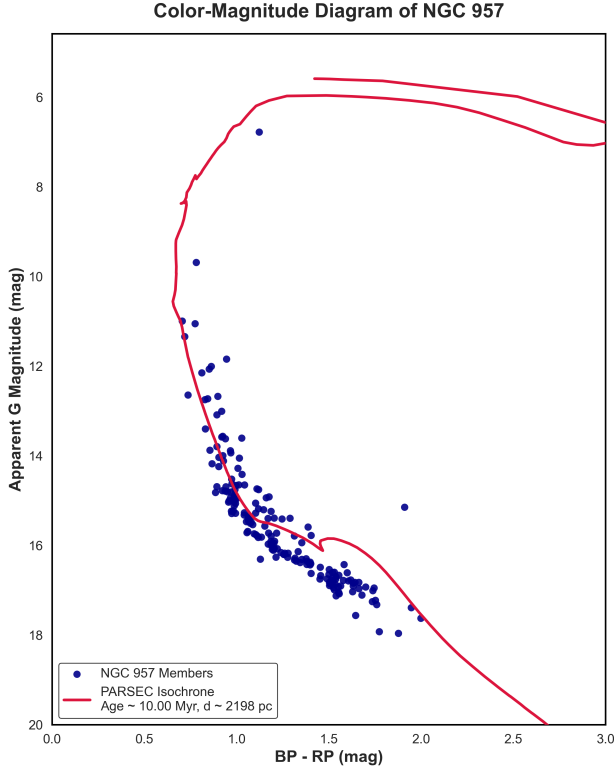


Figure 5: Gaia DR3 Colour-Magnitude Diagram for the astrometrically selected members of NGC 957. The PARSEC isochrone fit anchors the distance modulus, with the pre-main sequence transition zone visible at the faint end.

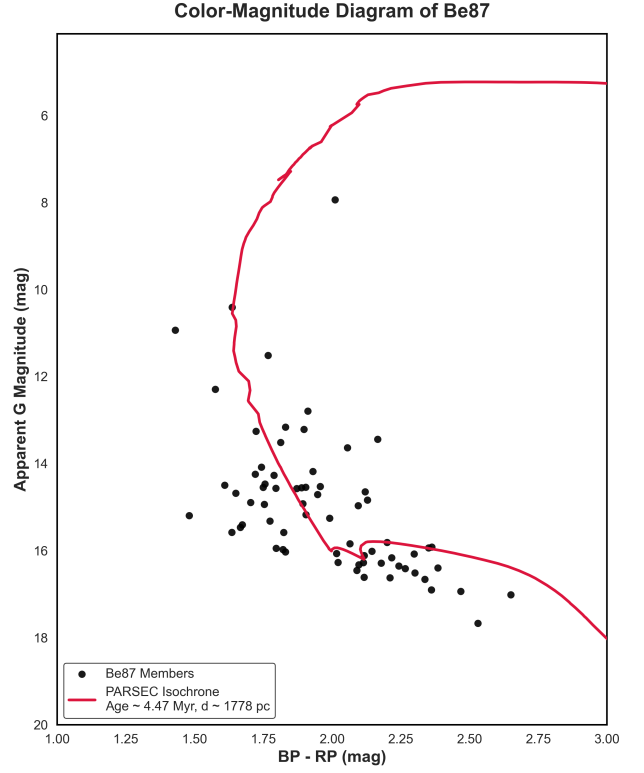


Figure 6: Gaia DR3 Colour-Magnitude Diagram for the astrometrically selected members of Be87, demonstrating the isolation of a distinct main sequence population against the extreme differential reddening of Cygnus X.

To validate the astrometric isolation, CMDs were constructed using the native Gaia DR3 passbands which can be seen in Figure 6 and 5. Theoretical stellar evolutionary tracks were generated from the PARSEC database [6] assuming a single solar metallicity ($Z = 0.0152$) for both clusters. The distance modulus for each cluster was anchored using the median zero-point-corrected Gaia DR3 parallax of the members [25]. Rather than relying on ages from the literature, these prior values were used to constrain the bounds of the theoretical isochrone generation. For Be87, tracks were generated over a wide grid between $\log(\text{Age}) = 6.30$ and 6.70 (2 to 5 Myr), while for NGC 957 they were generated between 6.60 and 7.05 (4 to 11 Myr), both utilizing a 0.05 logarithmic step. The derived distance of 2.2 kpc for NGC 957 is slightly larger than the Gaia-based literature average (2.0 kpc). Because the distance modulus is anchored directly to the median Gaia parallax, this slight discrepancy remains unexplained, but is independent of the isochrone metallicity.

A prominent widening is visible at the faint end of the main sequence ($G \gtrsim 15$) for both clusters. This feature is consistent with the Pre-Main Sequence (PMS) transition zone, though unresolved binaries and differential reddening also contribute to this spread [26, 2]. At ages of ~ 10 Myr or less, low-mass stars have not yet fully contracted

onto the Zero-Age Main Sequence (ZAMS) [63]. The PARSEC evolutionary tracks explicitly model this delayed contraction, which matches the rightward hook toward cooler temperatures observed in the Gaia photometry.

4.3.1. Isochrone Ages

The theoretical isochrone fit for NGC 957 yields an age of 10 Myr ($\log(\text{Age}) = 7.0$), which matches the literature mean (Table 1). For Be87, the updated astrometric isolation and clean main sequence allowed for an isochrone fit yielding an age of roughly 4.5 Myr ($\log(\text{Age}) = 6.65$). Decades of multi-wavelength observations establish Be87 as an extremely young embedded cluster, with literature ages ranging from 1 to 5 Myr and distances from 1.2 to 1.7 kpc [35, 13].

While our derived distance (1.80 kpc) differs from the historical literature mean, the derived age (4.47 Myr) naturally converged near the upper boundary of our constrained grid. This heavily constrained grid was guided by the physical presence of the Wolf-Rayet star WR 142 in the cluster region, which imposes a strict ≤ 5 Myr upper limit [37]. Although WR 142 was not recovered in the highly probable ($P \geq 0.9$) astrometric core, its native Gaia DR3 astrometry reveals why. WR 142 is excluded despite a consistent parallax and good RUWE, because it

moves just ~ 1.2 km/s relative to an arguably too-narrow proper-motion core (deviating by a Mahalanobis distance of $D_M \approx 3.65$). This shows the $P \geq 0.9$ plus 3σ truncation criterion trades completeness for purity, throwing out real, high-velocity members. Therefore, the age prior for the isochrone grid was derived strictly from the literature [35, 36].

Furthermore, an independent photometric survey utilizing the LCO 0.4m network placed H α emission constraints on the cluster. A lack of active circumstellar decretion disks was seen among the members in the FOV, including the known emission variable V439 Cyg, which restricts the current evolutionary state of the B-type population (Appendix A).

5. Conclusion

The primary objective of this work was to develop and validate a three-dimensional pipeline, **GaussCluster**, capable of isolating open clusters from crowded galactic environments. By operating strictly in a continuous probabilistic space ($\mu_{\alpha*}$, μ_{δ} , ϖ) via a Gaussian Mixture Model, the pipeline provides an alternative to density-based clustering algorithms, such as HDBSCAN, which rely on spatial density thresholds.

The software was validated on the open cluster NGC 957 and applied to the heavily obscured Cygnus X region to recover the Be87 cluster. The quantitative recovery metrics are summarized in Table 2.

Table 2: Membership overlap for Be87 and NGC 957 across the GaussCluster, STORM, and [27] catalogues, strictly restricted to a 15-arcminute radius for a homogeneous spatial comparison. Percentages express each intersection as a fraction of **GaussCluster**’s final output size for that cluster.

Catalogue / Pipeline	Berkeley 87	NGC 957
GaussCluster (3GMM)	70	180
STORM (15’)	146	311
Hunt & Reffert (15’)	82	252
GC $\cap$ STORM (%)	88.6%	91.7%
GC $\cap$ Hunt (%)	48.6%	74.4%

The overlap analysis reveals differences in membership allocation depending on the clustering architecture. The strong overlap with STORM ($\sim 90\%$) confirms the pipeline reliably anchors onto the initial dense core, while the moderate intersection (74.4%) with the [27] catalogue for NGC 957 highlights differences in spatial and astrometric limitations. While Hunt’s all-sky HDBSCAN methodology isolates a larger population, the 3GMM’s mathematical approach enables the probabilistic recovery of a more astrometrically restricted core. Comprehensive visual dashboards comparing our model against the catalogue of Hunt & Reffert (2023) for Berkeley 87, as well as against both

STORM and Hunt & Reffert (2023) for the validation cluster NGC 957, are available in Appendices B and C.

Photometric validation using Gaia DR3 passbands yielded distinct main sequences for both isolated astrometric groups. The isochrone fit for NGC 957 aligned strongly with the literature mean age of 10 Myr. Crucially, the fit for Be87 produced a young age of 4.5 Myr, which is consistent with the established 1–5 Myr age derived from decades of multi-wavelength observations [35, 13].

In summary, the **GaussCluster** pipeline provides a strict astrometric methodology for isolating cluster cores, while the photometric analysis demonstrates the ongoing need for complementary infrared observations or differential-reddening corrections when characterizing extremely young populations in the galactic plane.

6. Future Work

Future software development will focus on automating this two-stage process into a single, unified Graphical User Interface (GUI) or API wrapper. In this future release, users will simply specify a sky coordinate, allowing the software to trigger a STORM extraction (Stage 1) and pipe the output directly into **GaussCluster** (Stage 2) without requiring manual data-handling between the Python scripts.

Further observational work must also focus on obtaining deep near-infrared photometry (e.g., via the UKIRT Infrared Deep Sky Survey or JWST) and high-resolution spectroscopic follow-up for these newly confirmed astrometric members. Such observations are needed to characterise embedded populations [64], determine precise cluster ages [65], and assess the longevity of circumstellar disks [66].

By combining the blind discovery power of density searches with the astrometric precision of a 3GMM, this integrated tool will provide a framework for charting galactic open clusters.

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The computational framework and **GaussCluster** pipeline were built in Python, using its open-source ecosystem. This includes Astropy [67], NumPy [68], SciPy [69], pandas [70], scikit-learn [71], and Matplotlib [72].

Additionally, this research utilized the VizieR catalogue access tool and the SIMBAD astronomical database (operated at CDS, Strasbourg, France), alongside PARSEC theoretical stellar evolution models. Data retrieval used the Astroquery API [39].

Finally, this work has made use of data from the European Space Agency (ESA) mission *Gaia* (<https://www.>

cosmos.esa.int/gaia), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC, <https://www.cosmos.esa.int/web/gaia/dpac/consortium>). Funding for the DPAC has been provided by national institutions, in particular the institutions participating in the *Gaia* Multilateral Agreement.

Data Availability

The `GaussCluster` Python pipeline developed for this paper is open-source and publicly available at <https://github.com/raffey05-sudo/GaussCluster>. Furthermore, the complete reproducibility repository for this manuscript, containing the LaTeX source code, raw data catalogues, and all analytical scripts, is available at <https://github.com/raffey05-sudo/MSD-Dissertation>.

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A. H α Emission Constraints

To search for Classical Be-type stars and young stellar objects with circumstellar decretion disks, we conducted a targeted narrow-band H α photometric survey using the LCO 0.4m telescope network [73]. Due to the restricted Field of View (FOV) of the LCO imagers relative to the Gaia sky coverage, 34 of the 70 highly probable **GaussCluster** astrometric members physically fell within the footprint of the observations.

We utilized an automated aperture photometry pipeline to extract instrumental H α magnitudes for these 34 core members. By plotting the instrumental H α magnitude against the Gaia G_{RP} magnitude, we constructed an H α Diagram for the cluster (Figure 7). A 1:1 baseline was derived using the median offset of the cluster members, with a strict statistical boundary set at 3σ to flag emission-line outliers.

None of the 34 astrometric members crossed this formal 3σ emission threshold, remaining statistically indistinguishable from the baseline continuum. While the known peculiar emission-line variable V439 Cyg was recovered as a high-probability astrometric member, it did not exhibit significant emission excess during this observational epoch. This null result is consistent with the highly transient nature of Be-star decretion disks, which are known to periodically dissipate and reform over multi-year timescales [74]. The lack of active, strong circumstellar emission across the astrometric core places a meaningful physical constraint on the current evolutionary state of the cluster's B-type population.

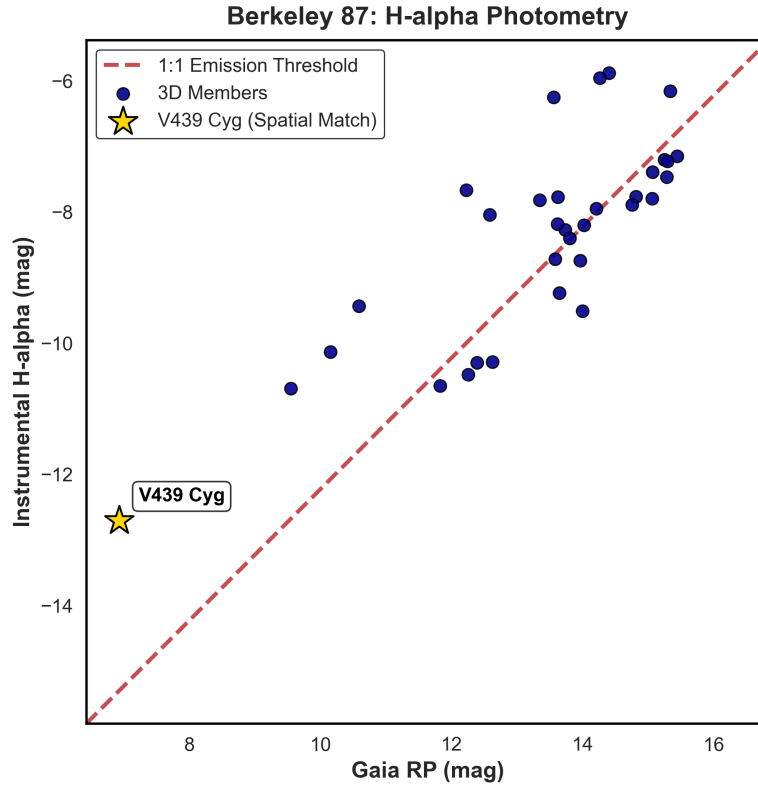


Figure 7: The H α Diagram for Be87. The 34 astrometrically confirmed core members within the FOV form the baseline. No members cross the formal 3σ statistical threshold, indicating a lack of active circumstellar decretion disks during this observational epoch.

B. Berkeley 87 Hunt Comparison

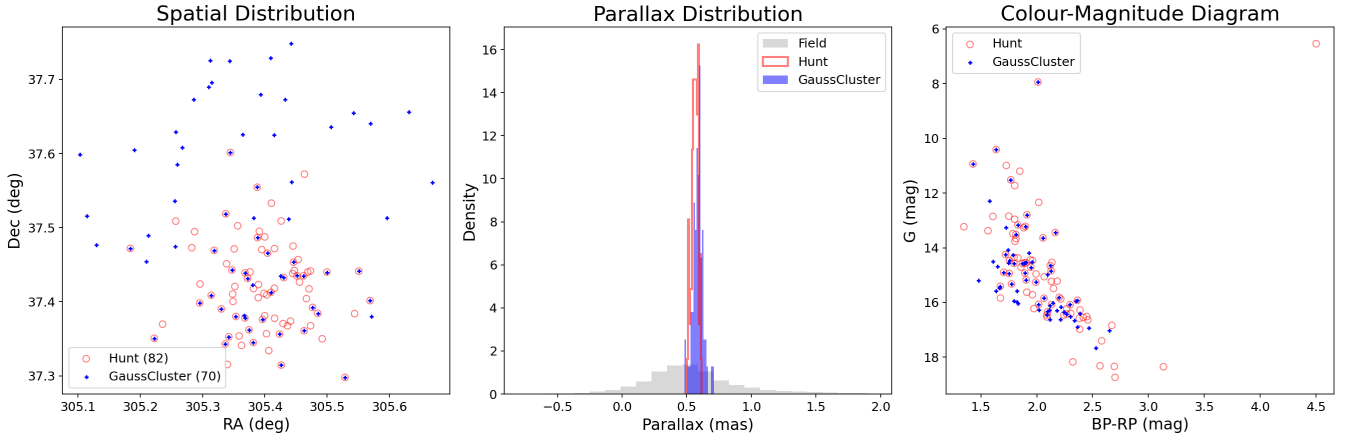


Figure 8: Spatial and parallax distribution comparison, along with a Colour-Magnitude Diagram for Berkeley 87 against the Hunt et al. [27] catalogue (red, 82 members within 15 arcminutes).

C. NGC 957 Validation Diagnostics

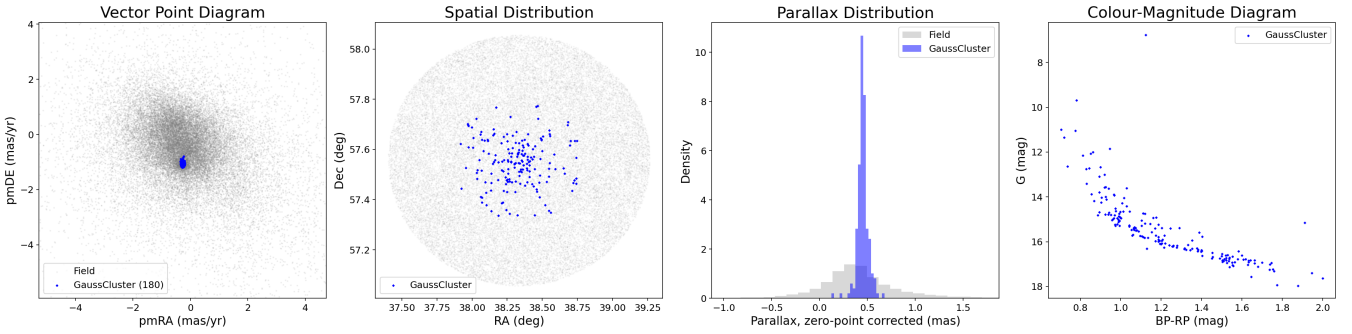


Figure 9: GaussCluster 4-panel diagnostic dashboard for NGC 957. The algorithm recovered 180 members (≥ 0.9), forming a tight Main Sequence.

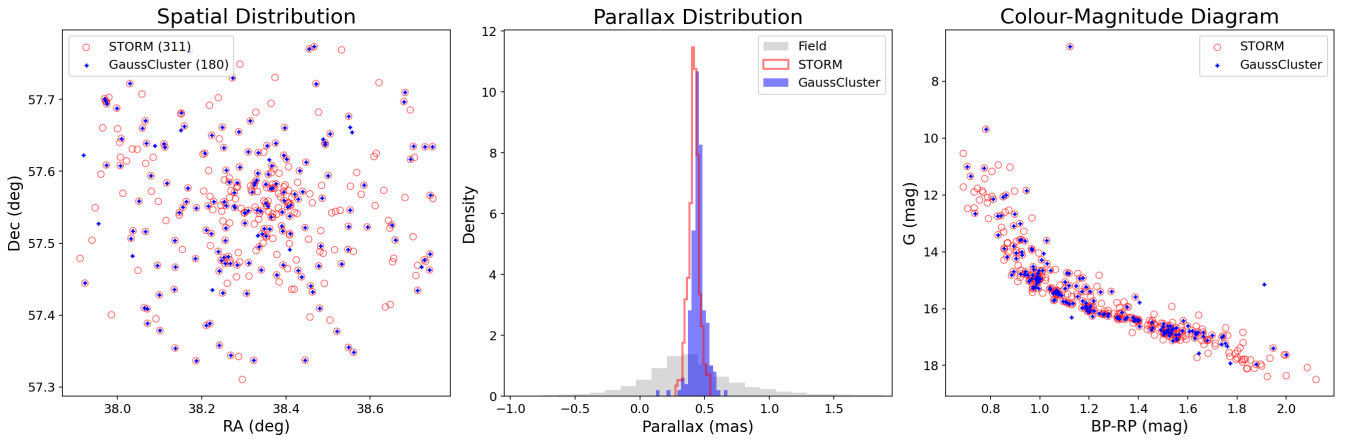


Figure 10: Spatial and parallax distribution comparison, along with a Colour-Magnitude Diagram for NGC 957. Our GaussCluster model (blue, 180 members) versus STORM (red, 311 members within 15 arcminutes).

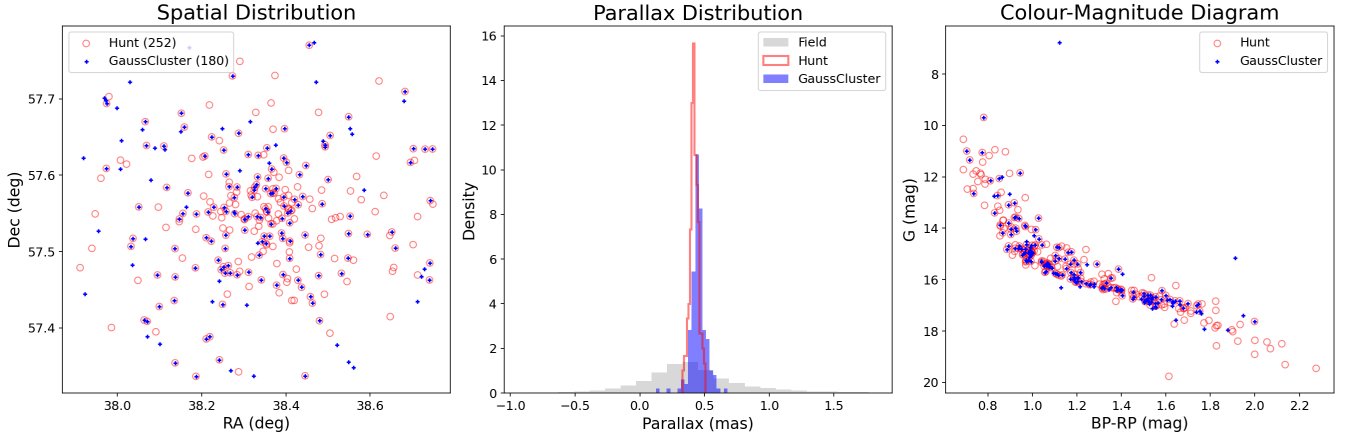


Figure 11: Spatial and parallax distribution comparison, along with a Colour-Magnitude Diagram for NGC 957. Our **GaussCluster** model (blue, 180 members) versus Hunt et al. [27] (red, 252 members within 15 arcminutes).

D. Spatial Verification (King Profiles)

To further validate the spatial structure of the astrometrically recovered members, we analyzed their radial surface density distribution. Open clusters exhibit a dense core surrounded by a diffuse halo, a structure mathematically well-described by an empirical King Profile [62]. By calculating the spatial density of the recovered members in concentric annuli expanding outward from the cluster centroid, we fitted standard King models to both Berkeley 87 and NGC 957 (Figure 12).

The resulting fits show a continuous decline in spatial density from the dense core towards the background field level. The recovered members exhibit a concentrated spatial distribution that is described by an empirical King profile. This provides a spatial consistency check on the astrometric membership selection, although the profile alone does not establish that the recovered population is gravitationally bound or relaxed. Notably, the astrometrically selected members closely trace the inner regions described by the King profile, despite spatial position not being included as an explicit dimension in the final **GaussCluster** membership model. This agreement provides additional evidence that the astrometrically selected population corresponds to the centrally concentrated component of the cluster, while sources with astrometric properties inconsistent with the fitted cluster distribution are excluded from the member sample. The agreement between the spatial and astrometric distributions therefore supports the interpretation of the recovered population as a cluster component, without requiring the King profile alone to establish its dynamical state.

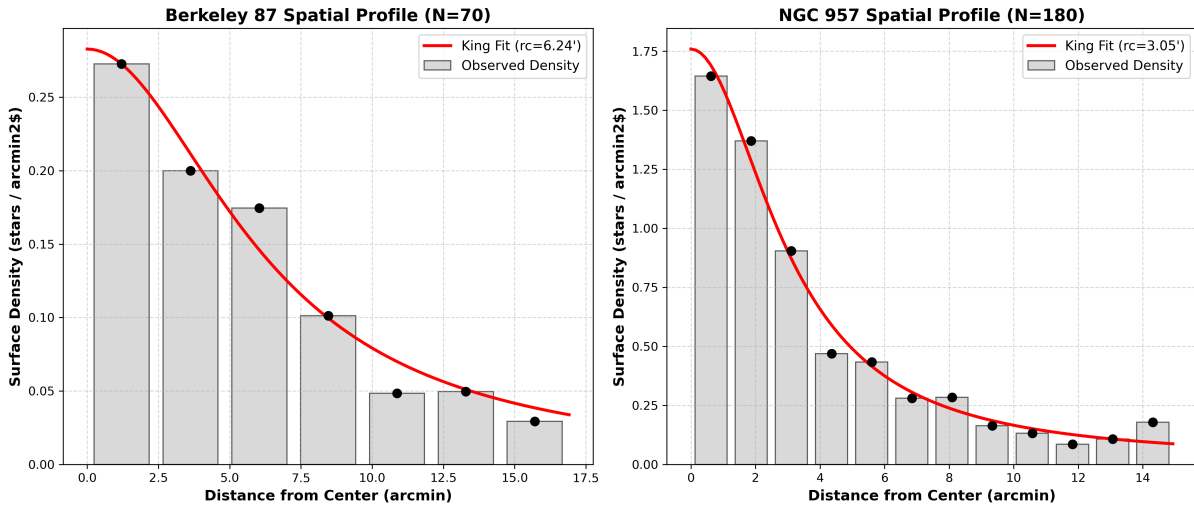


Figure 12: Empirical King Profile fits for the recovered spatial populations of Berkeley 87 and NGC 957. The steep decay to zero spatial density is consistent with the pipeline isolating the gravitationally bound inner core.

E. Berkeley 87 astrometric Catalogue

The complete catalogue of the 70 highly probable ($P \geq 0.9$) astrometric members of Berkeley 87, derived via the `GaussCluster` 3GMM pipeline is listed below in Table 3.

Table 3: Astrometric parameters and membership probabilities for the 70 astrometric members of Berkeley 87.

Gaia DR3 Source ID	RA (deg)	Dec (deg)	ϖ (mas)	$\mu_{\alpha*}$ (mas/yr)	μ_{δ} (mas/yr)	G (mag)	P_{memb}
2057967188004563968	305.4765	37.3923	0.544	-3.575	-6.185	13.22	0.97
2057966810047478272	305.4304	37.4325	0.562	-3.560	-6.193	15.26	0.97
2060973321517873024	305.2128	37.4891	0.531	-3.548	-6.182	14.50	0.97
2060974867706114176	305.2673	37.6081	0.565	-3.580	-6.095	14.08	0.97
2060976791851636224	305.2862	37.6729	0.527	-3.593	-6.234	15.58	0.97
2057966294651410560	305.3680	37.3780	0.554	-3.603	-6.145	15.18	0.97
2060995998936497664	305.5421	37.6546	0.537	-3.567	-6.237	13.80	0.97
2060981396056452864	305.1036	37.5984	0.573	-3.612	-6.147	15.20	0.97
2057991514699667840	305.5956	37.5132	0.609	-3.568	-6.183	16.27	0.97
2060967136764958592	305.2099	37.4545	0.596	-3.585	-6.182	16.29	0.96
2057965607456610176	305.4236	37.3563	0.556	-3.602	-6.274	14.47	0.96
2060969232708930816	305.3722	37.4314	0.557	-3.618	-6.201	14.57	0.96
2060971706610308352	305.4429	37.5617	0.627	-3.600	-6.212	15.95	0.96
2057960281697198592	305.3364	37.3428	0.576	-3.591	-6.293	14.90	0.96
2060971053775075584	305.3883	37.4865	0.590	-3.563	-6.276	15.81	0.96
2057968321875966592	305.4461	37.4534	0.562	-3.545	-6.283	14.65	0.96
2060999503642462976	305.4324	37.6729	0.582	-3.622	-6.183	15.58	0.96
2060974833346372352	305.2595	37.5849	0.577	-3.604	-6.078	15.47	0.96
2061000809309144064	305.3437	37.7247	0.547	-3.539	-6.083	14.18	0.96
2057967428522738432	305.4998	37.4397	0.602	-3.534	-6.228	12.62	0.96
2060975417462080768	305.3448	37.6014	0.549	-3.536	-6.254	15.84	0.96
2060977204168494720	305.3098	37.6895	0.537	-3.625	-6.113	13.26	0.95
2057966157212461056	305.3422	37.3526	0.523	-3.544	-6.247	16.46	0.95
2057994881954220800	305.6311	37.6562	0.560	-3.632	-6.136	14.53	0.95
2060980124746108032	305.1148	37.5158	0.549	-3.521	-6.205	16.12	0.95
2057993370125530240	305.6686	37.5612	0.539	-3.603	-6.321	14.55	0.95
2057967565961672576	305.5507	37.4412	0.566	-3.627	-6.252	15.92	0.95
2060969232708934784	305.3682	37.4389	0.551	-3.560	-6.341	13.44	0.95
2060970675817983488	305.3374	37.5186	0.558	-3.623	-6.287	14.56	0.95
2060970297860853760	305.3187	37.4694	0.566	-3.552	-6.341	13.16	0.95
2060969163989449216	305.3801	37.4224	0.497	-3.608	-6.285	16.40	0.95
2060963219754750208	305.2226	37.3506	0.590	-3.545	-6.324	15.41	0.95
2057966225931924992	305.3745	37.3620	0.574	-3.640	-6.222	15.32	0.94
2057964061268331648	305.5708	37.3800	0.496	-3.531	-6.275	16.36	0.94
2057966672608526464	305.4099	37.4126	0.588	-3.635	-6.255	16.07	0.94
2060977204168615040	305.3148	37.6960	0.492	-3.629	-6.179	16.90	0.94
2061000637510448896	305.4094	37.7288	0.569	-3.513	-6.127	15.94	0.94
2060967927038974080	305.1300	37.4765	0.588	-3.628	-6.277	16.66	0.94
2057962102763233792	305.5286	37.2979	0.553	-3.641	-6.245	14.25	0.94
2060976547029134080	305.2571	37.6288	0.566	-3.517	-6.283	12.30	0.94
2060968923471305856	305.2955	37.3982	0.578	-3.606	-6.353	14.58	0.94
2057965916694269312	305.3815	37.3448	0.553	-3.547	-6.361	10.94	0.94
2060971019415325568	305.4379	37.5115	0.449	-3.541	-6.292	16.63	0.93
2057965676176071808	305.4629	37.3608	0.531	-3.532	-6.334	14.55	0.93
2057967188004557952	305.4849	37.3843	0.512	-3.630	-6.307	14.92	0.93
2057968115717522560	305.4624	37.4350	0.578	-3.544	-6.362	12.80	0.93
2060971878408808192	305.3816	37.5128	0.648	-3.614	-6.080	17.02	0.93
2060995483549673088	305.5059	37.6361	0.509	-3.559	-6.017	16.52	0.93
2060972943560907392	305.4151	37.6248	0.531	-3.655	-6.195	16.03	0.93

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Table 3 – continued from previous page

Gaia DR3 Source ID	RA (deg)	Dec (deg)	ϖ (mas)	μ_{α^*} (mas/yr)	μ_{δ} (mas/yr)	G (mag)	P_{memb}
2060989058278236800	305.3122	37.7257	0.523	-3.548	-6.003	15.98	0.93
2060972084567248640	305.3878	37.5545	0.605	-3.651	-6.240	16.08	0.93
2057964920261827072	305.4264	37.3144	0.529	-3.612	-6.363	14.94	0.92
2057966874465505024	305.4255	37.4347	0.674	-3.511	-6.231	16.17	0.92
2060973012280208768	305.2564	37.4748	0.443	-3.560	-6.311	16.94	0.92
2057964507944938624	305.5687	37.4015	0.549	-3.581	-6.396	13.64	0.92
2060973871273685632	305.2557	37.5361	0.482	-3.643	-6.273	16.62	0.92
2060995724067830016	305.5696	37.6405	0.595	-3.627	-6.344	16.02	0.92
2057968081357787392	305.4516	37.4350	0.564	-3.616	-6.370	14.71	0.92
2060967411646651264	305.1841	37.4719	0.586	-3.506	-6.299	16.33	0.92
2060969370147881856	305.4042	37.4655	0.534	-3.628	-6.352	14.84	0.92
2060999641078034176	305.3933	37.6799	0.548	-3.592	-5.963	14.27	0.92
2061000293913064320	305.4423	37.7483	0.580	-3.506	-6.068	14.97	0.92
2060986786231472000	305.1902	37.6044	0.525	-3.543	-6.067	17.67	0.91
2060975795419202688	305.3647	37.6260	0.525	-3.658	-6.225	14.68	0.91
2060969404507635840	305.3472	37.4428	0.511	-3.661	-6.254	16.42	0.91
2057966290349918336	305.3665	37.3811	0.493	-3.495	-6.269	16.28	0.91
2060968682953119872	305.3302	37.3902	0.573	-3.574	-6.415	10.41	0.91
2060968614233634816	305.3535	37.3800	0.616	-3.631	-6.356	11.52	0.91
2057966049831160320	305.3963	37.3760	0.571	-3.659	-6.240	13.52	0.91
2060968957831040512	305.3140	37.4086	0.570	-3.567	-6.414	7.94	0.90